Projected collective measure: exact Wronskians, tail cancellation and a boundary-domain obstruction

Independent project audit

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Concrete result. The nonzero potential-tail cross terms do *not* change the finite-dimensional collective symplectic volume. Their contribution cancels from the collective Wronskian determinant, as verified directly on the exact two-soliton background during the collision **(c)**. There is nevertheless a precise obstruction to treating this result as a projected field determinant: the natural symplectic projector onto the full soliton family can send a localized fluctuation to a variation with nonzero potential endpoint. It therefore does not preserve a fluctuation domain with fixed potential endpoints **(c)**. A boundary-domain prescription and a finite-time determinant construction remain necessary **(d)**.

This continues `audit_determinant_foundations.tex`; it neither selects a field regularisation nor assigns an exchange phase. The repository good equation, canonical form $\Omega = \int d\pi \wedge d\phi$ and potential gauge $\phi(-\infty) = 0$ are retained. The calculations are for solution-parameter variations in that gauge. Every numbered formula carries a claim-(c) suffix and is checked in `scripts/audit_projected_measure.py`, with exact SymPy algebra and 55-digit mpmath quadrature. Numerical results are saved in `scripts/audit_projected_measure_results.json`.

1 The collective Wronskian has a factorised determinant

Let $a_j = \alpha_j$, $d_j = dP_j/da_j$, and order the tangent coordinates as $z = (a_1, a_2, X_1, X_2)$. For separated solitons with particle 1 on the left,

$$W_{AB} = \Omega(\partial_{z_A}, \partial_{z_B}) = \begin{pmatrix} C & D \\ -D & 0 \end{pmatrix}, \qquad D = \mathrm{diag}(d_1, d_2), \quad C = \begin{pmatrix} 0 & c \\ -c & 0 \end{pmatrix}, \quad c = -M_1'\Pi_2'. \tag{1c}$$

Here $M = 2\sin a$, $\Pi = -2\epsilon \sin a \cos a$ and $P = (2/3)\epsilon \sin^3 a \cos a$. The earlier audit directly checked the mixed entry by spatial integration. We now retain it in the full matrix. For arbitrary c, not only its Boussinesq value, exact matrix algebra gives

$$\det W = (d_1 d_2)^2, \qquad W^{-1} = \begin{pmatrix} 0 & -D^{-1} \\ D^{-1} & D^{-1} C D^{-1} \end{pmatrix}. \tag{2c}$$

Hence the positive Liouville density is

$$d\mu_{\mathrm{coll}} = |d_1 d_2|\, da_1\, da_2\, dX_1\, dX_2. \tag{3c}$$

There is no tail-dependent factor in this density. This resolves a specific concern raised in the earlier audit: individual mixed Wronskians need not vanish for their determinant to factorise.

The established Darboux coordinates make the cancellation explicit:

$$X_1' = X_1, \qquad X_2' = X_2 + \gamma_2 M_1, \qquad \gamma_2 = \Pi_2'/P_2'. \tag{4c}$$

Their tangent Jacobian is triangular with unit determinant. Pulling $\sum_j dP_j \wedge dX_j'$ back gives (1c); derivatives of γ_2 contribute only to a wedge with itself. The symbolic test verifies the full

matrix pullback, not merely its determinant. Switching the spatial ordering changes c and the corresponding shear, but leaves (2c) and (3c) unchanged. These formulas require $d_1 d_2 \neq 0$ and do not extend through the cusp by division.

2 Exact collision-time verification

Use the actual tau solution

$$\tau = 1 + E_1 + E_2 + A E_1 E_2, \quad E_j = e^{k_j(x - v_j t - x_{0j})}, \quad \phi = 2\partial_x \log \tau, \quad \pi = \partial_t \phi, \tag{5c}$$

with the repository's real KP cross-ratio $A > 0$. The script differentiates ϕ and π with respect to $(a_1, a_2, x_{01}, x_{02})$ analytically, then integrates every entry of their 4×4 Wronskian matrix. The Hirota parameters x_{0j} are not confused with both sets of physical asymptotic centers.

For $v_1 > v_2$, the incoming centers are $X_1^{\text{in}} = x_{01}$ and $X_2^{\text{in}} = x_{02} - \log A/k_2$; the outgoing ones are $X_1^{\text{out}} = x_{01} - \log A/k_1$ and $X_2^{\text{out}} = x_{02}$. Thus, in the fixed Hirota parameter basis, the mixed entry becomes

$$\begin{aligned} c_\tau &= \frac{d_2}{k_2}\partial_{a_1} \log A - M_1' \Pi_2' \\ &= -\frac{d_1}{k_1}\partial_{a_2} \log A + \Pi_1' M_2'. \end{aligned} \tag{6c}$$

The other blocks in (1c) remain D and zero. The equality of the two expressions is checked numerically along with the full spatial integral. At $(a_1, a_2, \epsilon_1, \epsilon_2) = (0.3, 0.7, +1, +1)$, all entries were checked at $t = -8, 0, 8$; the maximum discrepancy is below 1.3×10^{-55}. A head-on point $(0.25, 0.45, +1, -1)$ at $t = 0$ gives an error below 4.1×10^{-56}. In every case the determinant divided by $(d_1 d_2)^2$ is one to the stated precision **(c)**. These are observed residuals; the assertions require entry errors below 10^{-43} and a determinant-ratio error below 10^{-42}.

Why this is conserved despite the tails. For potential-field linearised solutions f, g, write $L = \partial_x^4 - \partial_x^2 + 6\partial_x u \partial_x$. A direct symbolic identity gives

$$\partial_x B(f, g) = fLg - gLf, \tag{7c}$$

$$B(f, g) = fg''' - f'g'' + f''g' - f'''g + (6u - 1)(fg' - gf'). \tag{8c}$$

Since $f_{tt} = -Lf$, the time derivative of $\int (f_t g - g_t f)dx$ is the endpoint difference of B. Soliton-family potential variations are bounded at spatial infinity and all their spatial derivatives decay there; therefore this boundary flux vanishes. The integrated Wronskian is conserved even though the amplitude variations themselves have constant tails. This supplies the missing direct justification for comparing the finite-time matrix to its asymptotic value. It proves a property of the finite collective solution space, not completeness of the continuum field modes.

3 An explicit symplectic projector and its limitation

Let Z_A denote the collective tangent solutions at a fixed time, and suppose a variation η belongs to a domain on which all pairings converge. Define

$$b_A = \Omega(Z_A, \eta), \qquad \mathcal{P}\eta = Z_A (W^{-1})^{AB} b_B, \qquad \mathcal{Q} = 1 - \mathcal{P}. \tag{9c}$$

Matrix multiplication gives

$$\mathcal{P}^2 = \mathcal{P}, \quad \mathcal{P} Z_A = Z_A, \quad \Omega(Z_A, \mathcal{Q}\eta) = 0. \tag{10c}$$

Replacing Z by ZR for an invertible collective-coordinate change leaves $\mathcal{P}$ unchanged, since W changes to $R^T W R$. The script verifies these identities in a six-dimensional canonical embedding with arbitrary d_1, d_2, c, including the Darboux shear. Thus tail cross terms can be incorporated in a concrete projector, rather than discarded.

A localized counterexample to domain preservation. Already for one soliton, take the localized initial-data perturbation $\eta = (\eta_\phi, \eta_\pi) = (0, u)$, where $u = (k^2/2)\,\mathrm{sech}^2[k(x - X)/2]$. Since $\phi_X = -u$ and $\phi_a(+\infty) = 2\cos a$, one finds

$$b_X = \int u^2 dx = \frac{2}{3}k^3, \tag{11c}$$

$$(\mathcal{P}\eta)^a = -\frac{2k^3}{3P'}, \qquad (\mathcal{P}\eta)_\phi(+\infty) = -\frac{4\cos a\, k^3}{3P'} \neq 0. \tag{12c}$$

The integral and projection are checked exactly. The original perturbation has zero potential variation at both endpoints; its projected part and its complement have compensating nonzero right tails. Consequently neither is in the same fixed-endpoint domain. This is an explicit obstruction, not merely the observation that ϕ_a is not square-integrable.

An enlarged space allowing such endpoint variations can accommodate (9c). A fixed-total-M space instead restricts the collective family. On that constraint,

$$da_2 = -\frac{M_1'}{M_2'} da_1, \qquad \ker \Omega\big|_{dM_{\mathrm{tot}}=0} = \mathrm{span}\left\{ \frac{M_1'}{P_1'}\partial_{X_1} + \frac{M_2'}{P_2'}\partial_{X_2} \right\}. \tag{13c}$$

The three-dimensional pullback has rank two, as checked symbolically. It therefore cannot simply inherit the four-dimensional measure (3c). Whether the quantum problem includes boundary variables, uses a constrained reduction, or has an equivalent scattering-state construction is not fixed by this calculation **(d)**. No such choice is made here.

4 What the measure result does not determine

Solution variations are not automatically determinant zero modes. The tangent fields solve the linearised equation, but they do not generally satisfy the time-endpoint conditions used to define a propagator determinant. A finite-dimensional benchmark makes the distinction precise. The free collective equation has solutions 1 and t, but neither is a nontrivial solution with zero values at both finite endpoints. For the elementary Dirichlet matrix $K_N = \mathrm{tridiag}(-1, 2, -1)$,

$$\det\left(\frac{m}{h}K_N\right) = (N+1)\left(\frac{m}{h}\right)^N \neq 0. \tag{14c}$$

Laplace expansion gives $D_N = 2D_{N-1} - D_{N-2}$ with $D_0 = 1, D_1 = 2$; substituting $D_N = N + 1$ proves the formula by induction. The recurrence and base cases are checked symbolically, alongside direct determinants for $N = 1, \ldots, 6$. This is a finite matrix diagnostic, not a proposed field regularisation. "Remove the zero modes" must specify which operator domain and which boundary problem are meant. A conserved solution Wronskian is not a proof that the finite-time operator has a kernel.

Volume and endpoint maps do not fix the propagation phase. A second exact control is the unit-frequency harmonic oscillator. Its classical symplectic flow returns to the identity at time 2π, preserves volume throughout, and has the same endpoint Jacobian as no evolution. But the ground-state wavefunction $e^{-x^2/2}$ satisfies $(-\partial_x^2 + x^2)\psi/2 = \psi/2$, so its phase after that evolution is $e^{-i\pi} = -1$ **(c)**. The script verifies both identities. Thus identical symplectic volumes and endpoint maps do not specify the phase carried by the evolution. This does not assign a Boussinesq Maslov index; it shows why (2c) cannot do so.

The remaining determinant task. The exact result now established is factorisation of the collective *symplectic* volume and of the determinant of its solution Wronskian. A Faddeev–Popov measure, a constrained Gaussian functional integral and a spectral determinant depend on more

data. One must give a common spatial and temporal domain, identify permissible collective directions on that domain, construct its complement or constraint Jacobian, and compare the interacting and one-body time evolutions with the same state normalization. The phase of the Gaussian square root must then be continued from an explicit reference. These are genuine outstanding field-theory requirements **(d)**. They cannot be supplied by choosing I or J because one satisfies an algebraic identity.

5 Checks, corrected derivatives and failures

For either direction, the signed momentum derivatives are

$$P'_{\text{good}}(a) = \frac{2}{3}\epsilon \sin a \sin 3a, \qquad P'_{\text{bad}}(b) = \frac{2}{3}\epsilon \sinh b \sinh 3b. \tag{15c}$$

Both are checked symbolically, including the factor ϵ omitted from formulas presented for both directions in the earlier report.

Result	Label	Check
Tail-independent collective Wronskian determinant and Darboux density	(c)	Exact symbolic matrices
Collision-time realization of that matrix	(c)	Four full spatial quadratures, 55 digits
Fourth-order symplectic boundary flux	(c)	Exact differential identity
Basis-independent collective projector	(c)	Exact canonical embedding
Fixed-endpoint obstruction and fixed-M rank loss	(c)	Explicit localized test and symbolic restriction
Boundary-condition and phase controls	(c)	Exact finite matrices and oscillator eigenstate
Projected field determinant and absolute phase	(d)	Not computed; domain and propagation data remain

Failures recorded. A PDF rebuild initially failed because removing a manual equation tag left a blank paragraph inside a display; removing that blank line fixed the build. The initial auxiliary six-dimensional canonical embedding had the opposite sign, producing $Z^T J Z = -W$ instead of W. The exact matrix check caught this; the embedding signs were corrected. No tolerance was changed. No collision quadrature failed. The fixed-endpoint failure of the projector is an intended audit finding, not a numerical failure. The earlier sign omission in the momentum derivatives is corrected explicitly above.

References and provenance. This note uses the repository's tau solution and prior classical symplectic form as stated conventions and independently checks the needed derivatives, Wronskians and matrix identities. It adds no external literature claims or new references. The preceding determinant audit records the extent of the FK source passages actually read.